Supporting Information for

Supporting Information for: Systematic dissection of endogenous A-to-I RNA editing reveals substrate- and target-level determinants of predictability

Author list pending

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Supporting Information Text

This appendix provides the complete scientific argument behind the ADARPro campaign, including branches that were limited, stopped, or deferred. Four evidence categories remain distinct throughout: *development OOF* denotes pair-disjoint cross-fitted comparisons used for discovery and selection; *selection diagnostic* denotes published-validation quantities used for checkpoints or thresholds; *canonical held-out* denotes the single evaluation after the final manifest was frozen; and *secondary* denotes robustness or descriptive analyses that do not define the benchmark claim. "Not supported" always means not supported by the tested representation at the available resolution. A deferred hypothesis lacked a benchmark-compatible measurement and was not counted as negative evidence.

A. Benchmark reproduction and leakage control

Immutable AdarEdit benchmark. The upstream AdarEdit reproducibility release was preserved at the immutable revision recorded in repository provenance. Environment, package, GPU, input-file, and checkpoint provenance are retained in the repository. The upstream verifier recovered 884 graph-buildable inverted-Alu pairs: 566 train, 139 validation, and 179 test, with no pair shared between partitions. Supplied Baseline GAT checkpoints reproduced the reported metrics for artery tibial, brain/cerebellum, liver, skeletal muscle, and Combined. Combined pools evidence across GTEx tissues and was never treated as a fifth tissue when constructing environmental covariates.

Each example contains both Alu arms in transcript orientation, a focal adenosine, predicted dot-bracket structure, and EditingIndex-derived label. EditingIndex $\geq 15\%$ defines edited sites, EditingIndex $< 1\%$ defines nonedited sites, and the canonical binary task excludes the intermediate interval. Nodes encode nucleotide and structure-derived features; edges represent backbone and predicted base-pair relations. The original construction used ViennaRNA for secondary structure (1).

Coordinates and authoritative observations. Genomic coordinates were reconstructed with the original strand-aware transformation from local model positions to *hg38*/GRCh38. Pair intervals came from `Pair_Alu_withStrand.bed`; raw tissue tables supplied focal positions and EditingIndex. Both `site_id = chromosome:position:strand` and `(pair_id, local_position)` were retained. Quality control required an adenosine in transcript orientation, deterministic row mapping, correct strand, and stable identity across tissues. The resulting `site_master.parquet` records labels, continuous editing values, sequence, structure, coordinates, folds, OOF predictions, and provenance.

Test quarantine, grouped folds, and inference. Canonical test paths and SHA-256 values were registered before development. The quarantine library denied protected-path access unless a single-use authorization matched the frozen manifest. Discovery used the union of published train and validation pairs divided deterministically into 5 outer folds by pair ID. Preprocessing, feature selection, dimensionality reduction, fitted residual models, checkpoints, and thresholds excluded the outer fold. A row entered an OOF table only after a model that had not seen its pair scored it.

The grouped OOF reference is an internal research instrument, not a revision of the published benchmark. Each outer model sees fewer pairs than a final fit and therefore has lower absolute performance. OOF comparisons estimate incremental value because candidate and reference share folds, training scope, and evaluation logic. The supplied-checkpoint canonical comparison answers the final benchmark question (Fig. S1). Development differences are never added arithmetically to published test scores.

We report AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, Matthews correlation coefficient, and Brier score. Unless specified otherwise, thresholds maximized F1 on fold-internal validation data. Paired confidence intervals use 2,000 bootstrap replicates sampling pair IDs with replacement and retaining every site from a sampled pair. Effect size and clustered uncertainty, rather than site-wise P values, determine interpretation.

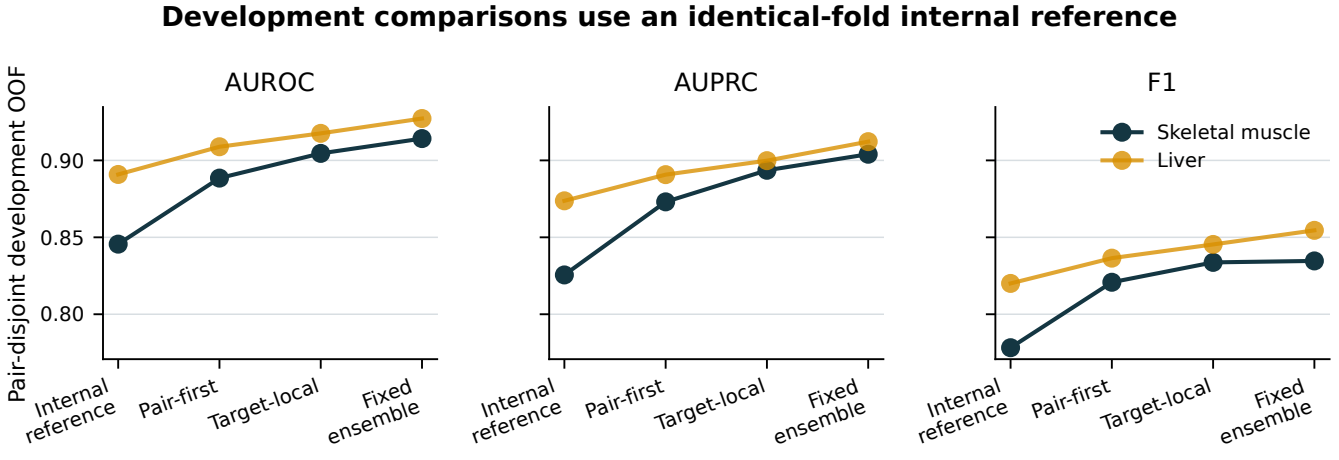


Fig. S1. Two evaluation regimes answer different questions. Development OOF comparisons use an identical-fold internal reference to attribute incremental effects. The separately reproduced supplied-checkpoint canonical test is the authoritative final benchmark.

Table S1. Complete one-use canonical held-out metrics. Thresholds were frozen from the published validation partition before test access. Combined is a pooled endpoint.

Context	Model	AUROC	AUPRC	F1	Precision	Recall	Specificity	Brier
Artery	AdarEdit	0.9238	0.9262	0.8671	0.8345	0.9023	0.7930	0.1077
Artery	ADARPro	0.9477	0.9485	0.8869	0.8840	0.8899	0.8650	0.0879
Brain	AdarEdit	0.9301	0.9286	0.8689	0.8371	0.9032	0.8021	0.1050
Brain	ADARPro	0.9508	0.9504	0.8902	0.8741	0.9069	0.8529	0.0855
Liver	AdarEdit	0.9092	0.9242	0.8541	0.8368	0.8721	0.7788	0.1196
Liver	ADARPro	0.9414	0.9500	0.8812	0.8946	0.8683	0.8670	0.0951
Skeletal muscle	AdarEdit	0.8688	0.8757	0.8139	0.7767	0.8549	0.7152	0.1526
Skeletal muscle	ADARPro	0.9144	0.9207	0.8541	0.8230	0.8876	0.7788	0.1296
Combined	AdarEdit	0.9331	0.9356	0.8674	0.8334	0.9043	0.8026	0.1043
Combined	ADARPro	0.9491	0.9513	0.8852	0.8463	0.9279	0.8159	0.0870

B. Complete hypothesis map

The research tree was ranked before expensive model development by mechanistic rationale, novelty relative to AdarEdit, data compatibility, leakage risk, effective sample size, and expected performance upside. Directions advanced only after comparison with the appropriate same-fold reference and negative controls. The resulting taxonomy separates selected contributions, limited evidence, unsupported tested formulations, and hypotheses deferred because the required measurement was unavailable (Table S2). The complete machine-readable tree records each hypothesis, source, promotion rule, kill rule, and final decision; Table S10 preserves the chronological disposition without turning run counts into a scientific result.

Table S2. Development evidence map. "Not supported" refers only to the tested measurement and formulation; deferred hypotheses lacked appropriately matched data and were not tested. Quantitative comparisons and their prespecified references are reported in the corresponding sections and source tables.

Scale	Tested formulation	Evidence status	Interpretation
Substrate-aware learning	Pair-first expanded supervision	Selected	Corrects label use and pair weighting
Target selectivity	Aligned target + helix-partner view	Selected	Survives shifted-window control
Integration	Fixed equal-logit global + local	Selected	Complementary same-fold predictions
Substrate realization	Pair geometry / editing burden	Supported	State is measurable, not yet actionable
Supervision	Continuous + ordinal auxiliary labels	Limited	Ranking benefit; no useful F1 gain
Target selectivity	Focal node / contact geometry	Limited	Points toward aligned local view
Cross-tissue	Shared four-tissue training	Limited	Transfer below promotion gate
Substrate realization	Hierarchical engagement gate	Limited	Small ranking gain; unstable classification
Biological context	Bulk H3K36me3 overlap	Limited	Context-specific AUPRC only
Biological context	RBP binding × bulk expression	Not Supported	No reproducible residual gain
Biological context	Transcript / repeat annotation panel	Not Supported	No robust site-level increment
Representation	Frozen RNA-language embeddings	Not Supported	No consistent incremental utility
Representation	Substrate-wide all-A decoder	Not Supported	Stopped at prespecified futility gate
Capacity	Larger / deeper graph networks	Not Supported	Capacity alone did not generalize
Integration	Unified end-to-end architecture	Not Supported	Inferior to frozen ensemble
Cellular state	Matched donor/cell multimodality	Deferred	Required measurements unavailable
RNA state	Matched in-vivo structure / isoforms	Deferred	Required measurements unavailable

C. Biological-context tests

RBP occupancy, binding potential, and abundance. Candidate RBPs were prioritized independently from model outcome using known ADAR interaction, dsRNA binding, perturbation evidence, nuclear localization, Alu association, expression, and assay

availability. Experimental occupancy used replicated or high-confidence ENCODE eCLIP peaks (2); predicted occupancy used PrismNet and RBPNet where compatible (3, 4). Features included focal overlap, distance to the nearest peak, local-window signal, duplex-wide burden, arm asymmetry, partner occupancy, and regulator-class aggregates. GTEx medians supplied tissue abundance and binding-by-abundance interactions (5). Combined received no artificial tissue vector.

Controls included randomized RBP identities, shuffled tissue vectors, matched-dimensional noise, abundance without binding, and binding without abundance. High-confidence baseline false positives were matched on baseline probability, tissue, structural state, and pair characteristics. None of the experimental-occupancy, predicted-binding, abundance, or interaction panels produced a reproducible residual increment beyond its controls (Fig. S2). This operationalization is limited by cell-line/tissue mismatch and repeat mapping; absence of an eCLIP peak in Alu sequence was never interpreted as biological absence.

Label-switching sites hold sequence and predicted structure constant across individual tissues. In the available four-tissue comparison, tissue identity nearly separated switch direction by itself, so tissue-average expression could not establish a regulator-specific transferable mechanism. The branch was terminated rather than enlarged. Established RBP regulation of editing remains biologically compatible with this predictive result (6–9).

Transcript, repeat, and chromatin annotations. Reconstructed pair intervals supplied arm separation and length, orientation, repeat subfamily and divergence, Alu-neighborhood composition, transcript overlap, and exon/intron annotations. Fold-isolated residual models combined standardized features with the baseline logit. The broad site-level panel did not generalize beyond the structure graph; this did not preclude a pair-level role for geometry (Section D).

Exact-tissue compatible H3K36me3 tracks were summarized at the target, local windows, arms, and entire pair. A chromosome-shifted overlap acted as a spatial control. H3K36me3 produced a weak liver AUPRC effect but no stable AUROC, F1, or engagement improvement (Fig. S2). H2A.Z was not promoted to a benchmark experiment because no adequately matched and powered tissue resource was available. The result concerns information in bulk peak overlap, not chromatin causality (10, 11).

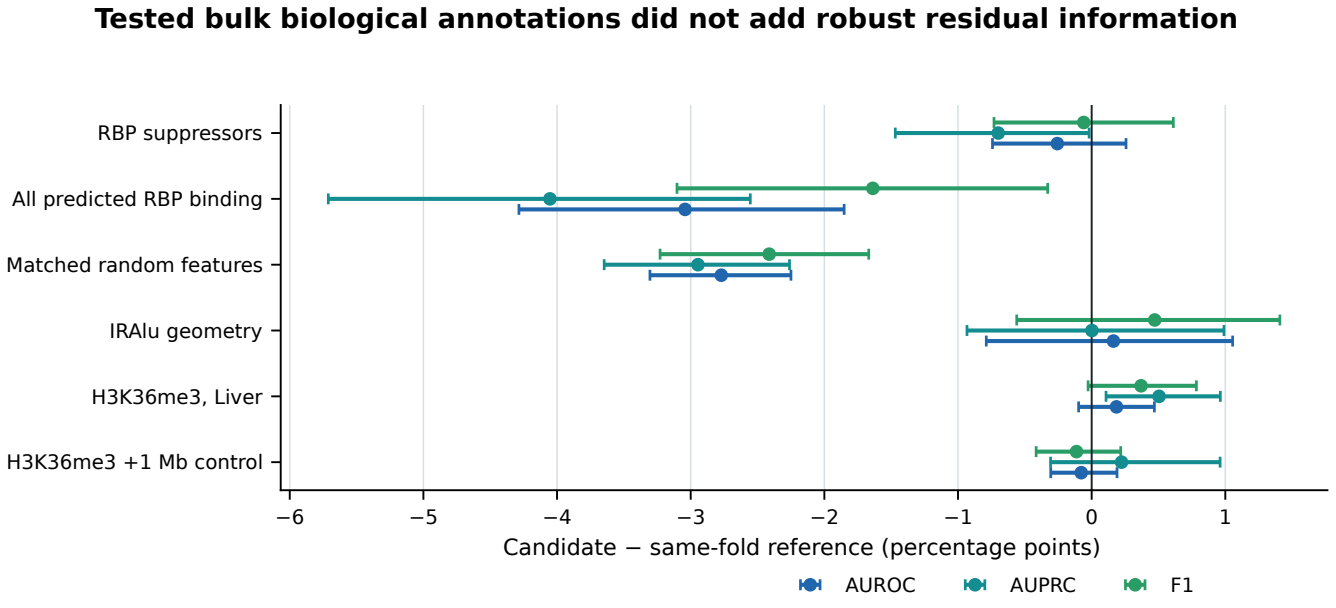


Fig. S2. Bulk biological annotations and matched controls. Every point is labeled by formulation. Pair-bootstrap development differences use the documented same-fold reference. The RBP, genomic, and chromatin measurements shown did not add a reproducible increment across ranking and thresholded metrics.

D. Substrate realization

Pair-level analyses used one row per development substrate. Outcomes were edited-site fraction, mean EditingIndex, and whether a pair contained any high-editing site. Predictors ranged from adenosine count and graph-derived sequence/structure summaries to genomic geometry and the combined 16-feature low-capacity model. All fitting remained inside pair folds.

Inter-arm gap correlated negatively with edited-site fraction in every individual tissue. Geometry plus graph summaries improved R^2 over graph summaries alone in all four tissues, with positive pair-bootstrap intervals (Table S3). The combined model also classified whether a pair contained any high-editing site above chance. These are direct observations that substrates differ in realized burden; they do not identify the cellular mechanism that realizes a duplex.

Site-level translation was weaker. Multiplying or gating site probabilities with cross-fitted engagement estimates increased AUPRC modestly but did not yield a stable F1 gain and worsened Brier score. A neural auxiliary engagement objective decreased the principal classification metric and was stopped. Static pair attributes can therefore describe realization propensity without recovering the time-, isoform-, transcript-, or cell-specific state required to alter the strongest site predictions.

Table S3. Substrate realization is measurable in development pairs. Spearman ρ relates genomic inter-arm gap to edited-site fraction. The low-capacity pair model combines genomic geometry with graph-derived summaries; ΔR^2 is relative to graph summaries alone with a 95% whole-pair bootstrap interval. "Any high" denotes whether a pair contains at least one site with EditingIndex $\geq 15\%$.

Context	Gap–burden ρ	Graph-only R^2	Combined R^2	ΔR^2 [95% CI], points	Any-high AUROC
Artery	-0.482	0.053	0.270	+21.64 [+16.39, +26.93]	0.816
Brain	-0.425	0.050	0.239	+19.11 [+13.94, +23.68]	0.749
Liver	-0.475	0.070	0.277	+20.82 [+15.75, +26.03]	0.728
Skeletal muscle	-0.334	0.010	0.050	+4.12 [+0.99, +6.98]	0.705

E. Richer supervision and the statistical unit

The expanded training pool contains all eligible high- and low-editing observations in each training pair. The site-balanced formulation draws uniformly from those rows. The pair-first formulation draws a pair uniformly, a class with probability 0.5, and then an eligible site of that class; if the pair lacks that class, its available class is used. The canonical-row epoch budget was held fixed, preventing extra optimizer steps from explaining the difference.

The existing same-fold sequence original reference → expanded site-balanced → expanded pair-first separates the two corrections (Table S4). Label recovery improved all three primary metrics. With the label pool held constant, pair-first weighting added a second favorable increment, including +2.85 F1 points. Pair-first gains replicated across all endpoints (Table S5). This supports the pair as the unit of statistical dependence, without asserting that every site within a pair has identical biology.

The binary/ordinal/continuous model shared an encoder across canonical classification, ordinal bins, and transformed EditingIndex regression. It added a small ranking and Brier improvement over pair-balanced binary training, but essentially no F1. Adding it to the selected predictor reduced natural-prevalence F1, so the graded-label branch remained limited. Intermediate values contain ordering information, but they did not improve the primary thresholded task enough to justify added complexity.

Table S4. Same-fold decomposition of expanded supervision and pair-first weighting in muscle development data. Intervals are whole-pair bootstrap intervals for the incremental step indicated, in percentage points.

Metric	Original reference	All high/low, site-balanced	Δ from labels [95% CI]	All high/low, pair-first	Δ from pair weighting [95% CI]
AUROC	0.8456	0.8659	+2.04 [+1.06, +3.00]	0.8885	+2.27 [+1.25, +3.36]
AUPRC	0.8256	0.8520	+2.63 [+1.25, +4.17]	0.8730	+2.08 [+0.76, +3.55]
F1	0.7783	0.7924	+1.42 [+0.21, +2.57]	0.8208	+2.85 [+1.68, +4.01]

Table S5. Pair-first expanded supervision versus the same-fold internal AdarEdit reference. Values are percentage-point median differences with 95% whole-pair bootstrap intervals.

Context	ΔAUROC [95% CI]	ΔAUPRC [95% CI]	ΔF1 [95% CI]
Artery	+1.22 [+0.80, +1.66]	+1.37 [+0.79, +1.97]	+0.69 [+0.10, +1.27]
Brain	+1.17 [+0.82, +1.54]	+1.64 [+1.02, +2.20]	+1.11 [+0.61, +1.64]
Liver	+1.79 [+1.18, +2.37]	+1.72 [+0.70, +2.71]	+1.64 [+0.96, +2.30]
Skeletal muscle	+4.28 [+3.07, +5.49]	+4.73 [+3.15, +6.35]	+4.23 [+3.11, +5.38]
Combined	+1.45 [+1.09, +1.84]	+1.69 [+1.15, +2.24]	+1.41 [+0.94, +1.98]

F. Focal and target-local representation

Three stages tested whether whole-duplex mean pooling loses site identity. First, fold-isolated target-contact geometry added AUROC and AUPRC but little F1, localizing a ranking effect near the target. Second, a focal-node graph model preserved the final target embedding alongside the global mean. It improved ranking relative to the pair model and an otherwise matched duplicate-global control, although its standalone F1 interval was insufficient for promotion. Third, the selected 65-position encoder aligned the focal adenosine with its dot-bracket partner and encoded nucleotide, pairing, validity, target/partner indicators, and relative position through two dilated convolutional layers.

The aligned local model exceeded the whole-duplex pair model in muscle and replicated in liver. Its matched-capacity control moved the same local window 96 nt away while retaining all dimensions and optimization. The shifted model collapsed across AUROC, AUPRC, and F1 (Table S6), demonstrating that alignment rather than generic convolutional capacity carries the improvement. Natural-prevalence results preserved the advantage (Fig. S3).

Table S6. Target-representation controls in muscle development OOF data. All values are evaluated on the same canonical development rows.

Representation	AUROC	AUPRC	F1	Brier
Whole-duplex pair model	0.8885	0.8730	0.8208	0.1372
Explicit focal node	0.9014	0.8902	0.8251	0.1342
Aligned target + partner	0.9045	0.8936	0.8337	0.1286
Matched-capacity +96-nt shift	0.7122	0.6943	0.7028	0.2356

G. Cross-tissue sharing

A four-output masked model shared a graph encoder across individual tissues while omitting unobserved labels from the loss. Relative to independently trained pair-balanced models, it yielded limited positive transfer, concentrated in the data-limited skeletal-muscle endpoint. Macro AUROC/AUPRC/F1 changes were 0.0087/0.0114/0.0075, but the prespecified AUROC and F1 promotion gates were not both met and mean Brier increased by 0.0043. Transferable intrinsic rules therefore exist, but their effect was smaller and less calibrated than the improvements from supervision and target alignment. Combined was excluded from this biological multitissue formulation.

H. Foundation-model and substrate-wide experiments

Frozen RiNALMo representations were evaluated at the target and substrate scales, including a larger frozen variant (12). Residual models and neural fusion used fold-fitted normalization and matched-dimensional controls. Neither variant added consistent AUROC/AUPRC beyond the selected graph representation. This is evidence against generic frozen embeddings as an additive feature in this benchmark, not against RNA pretraining. A corpus audit found 288,056 benchmark-excluded opposite-orientation Alu pairs in GENCODE transcript spans, but transcript connectivity, foldability, and sequence-redundancy filters were insufficiently resolved for benchmark-safe task-specific pretraining. That direction remains deferred.

The substrate-wide decoder encoded a pair once and classified all candidate adenosines jointly. It was evaluated against a pair-balanced model on identical rows with a prespecified two-fold futility rule. The decoder lost -0.0664 AUROC and -0.0599 F1, with wholly adverse intervals, and training stopped. A relative within-pair rank feature helped a weaker original encoder but neither beat pair-balanced training nor its shuffled-rank control. Multiple sites provide useful supervision; the tested all-A decoder did not exploit it better than pair-first stochastic learning.

I. Natural-prevalence robustness

The canonical benchmark is balanced for direct comparison, so its precision is not a deployment-prevalence estimate. The secondary natural-prevalence evaluation retained every eligible high/low observation from unchanged development pairs. In both muscle and liver, the fixed ensemble improved ranking, AUPRC, F1, and false-positive count over its pair-balanced component (Fig. S3; Table S7). These results support practical false-positive reduction but do not replace the canonical task.

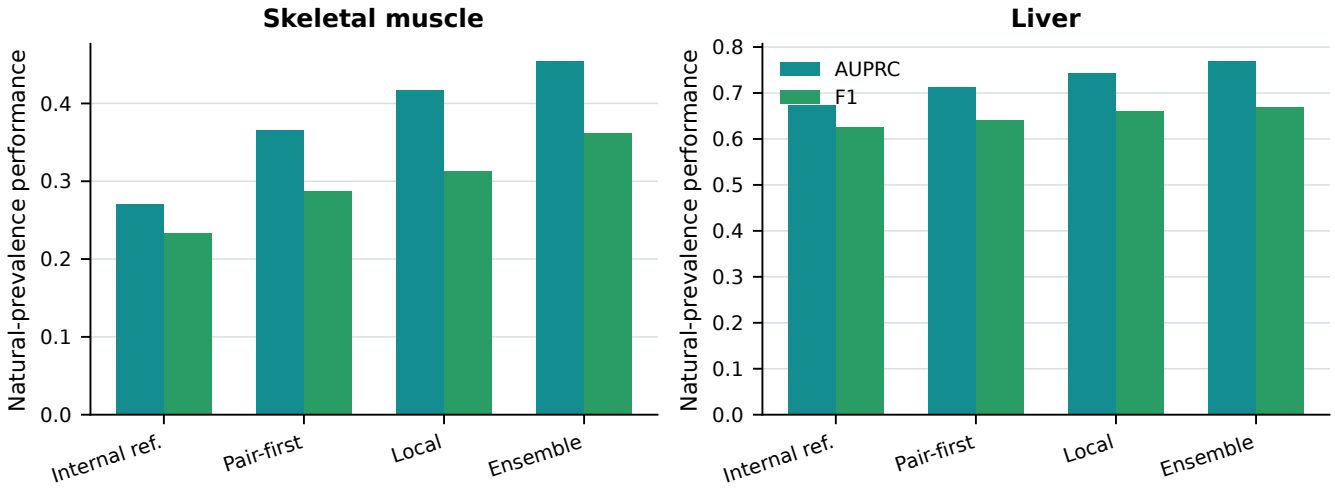


Fig. S3. Secondary natural-prevalence development evaluation. Performance for the same-fold internal reference, pair-first whole-duplex model, target-local component, and fixed ensemble on all eligible high/low observations from unchanged muscle and liver development pairs.

Table S7. Secondary natural-prevalence development evaluation. All eligible high/low observations from unchanged development pairs are included; fold-specific validation thresholds are used.

Context	Model	n	AUROC	AUPRC	F1	Precision	Recall	FP
Skeletal muscle	Internal reference	54289	0.8478	0.2711	0.2328	0.1351	0.8415	15804
Skeletal muscle	Pair-balanced	54289	0.8900	0.3659	0.2868	0.1714	0.8773	12441
Skeletal muscle	Target-local	54289	0.9072	0.4166	0.3125	0.1900	0.8800	11007
Skeletal muscle	Fixed ensemble	54289	0.9165	0.4540	0.3621	0.2306	0.8422	8245
Liver	Internal reference	38096	0.8915	0.6732	0.6246	0.4906	0.8592	6690
Liver	Pair-balanced	38096	0.9094	0.7128	0.6418	0.5056	0.8785	6442
Liver	Target-local	38096	0.9200	0.7432	0.6602	0.5265	0.8851	5969
Liver	Fixed ensemble	38096	0.9290	0.7687	0.6684	0.5306	0.9028	5988

J. Preregistered unified end-to-end experiment

After the prediction-level ensemble was frozen but before test access, a separate protocol specified one connected model. A shared three-layer graph attention encoder produced whole-duplex and focal-node states; the aligned local CNN supplied the third view. The three projected representations and their pairwise elementwise interactions entered a 64-unit fusion layer, while a training-only substrate head carried fixed loss weight 0.25. The connected model had 65,698 trainable parameters versus 80,138 parameters across the two independent ensemble components. Training used pair-first class balance, 1,000 epochs, Adam at learning rate 10^{-3} and weight decay 10^{-4}, batch size 256, and seed 20,260,821. The comparator prediction hash and staged decision rules were preregistered.

The unified model's AUROC and AUPRC were below the frozen ensemble in every outer fold (Fig. S4; Table S8). Aggregate F1 was similar, but ranking intervals were entirely adverse and calibration was substantially worse. The conditional no-auxiliary repair, liver replication, and all-context stages were therefore not opened. This preserves the decision rule and shows that independently trained representations were more reliably combined at prediction level than through the tested joint optimization.

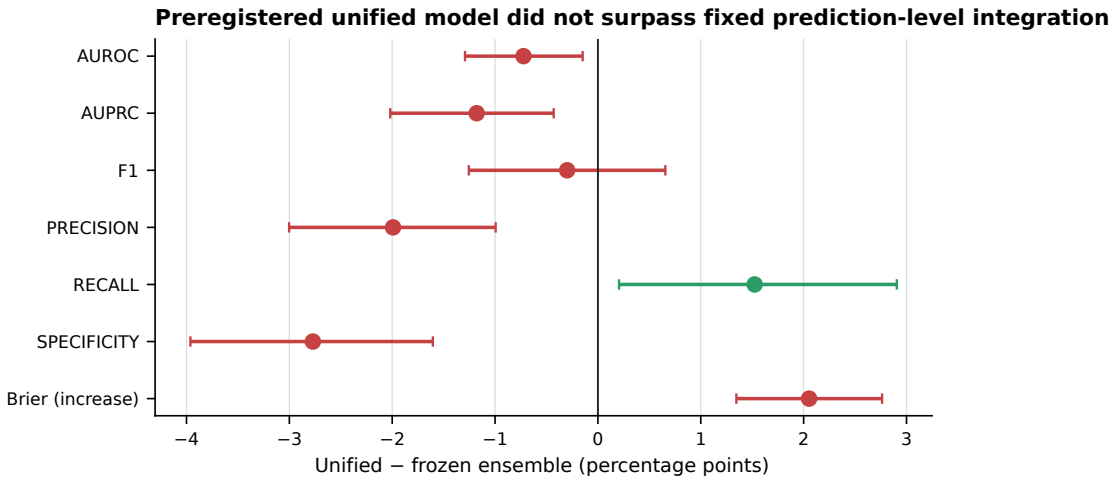


Fig. S4. Preregistered unified model versus the immutable ensemble. Bars are 95% whole-pair bootstrap intervals for unified minus ensemble in muscle development OOF data. A positive Brier difference is adverse.

Table S8. Preregistered unified-model comparison on muscle development data. The unified architecture was rejected before canonical-test access.

Model	AUROC	AUPRC	F1	Precision	Recall	Specificity	Brier
Internal reference	0.8456	0.8256	0.7783	0.7238	0.8415	0.6894	0.1695
Unified end-to-end	0.9072	0.8922	0.8317	0.8074	0.8575	0.8022	0.1372
Frozen ensemble	0.9142	0.9040	0.8347	0.8273	0.8422	0.8299	0.1168
Target-local	0.9045	0.8936	0.8337	0.7920	0.8800	0.7765	0.1286

Metric	Unified − ensemble, points [95% CI]
AUROC	-0.72 [-1.29, -0.15]
AUPRC	-1.18 [-2.02, -0.43]
F1	-0.30 [-1.25, +0.66]
PRECISION	-1.99 [-3.00, -0.99]
RECALL	+1.52 [+0.21, +2.91]
SPECIFICITY	-2.77 [-3.96, -1.60]
BRIER	+2.05 [+1.35, +2.76]

K. Final training and one-use canonical evaluation

For each endpoint, pair-first whole-duplex and aligned-local components were fitted on the published training pairs. Published validation data selected checkpoints and one ensemble threshold per endpoint. Ten checkpoints, histories, configurations, selected epochs, thresholds, inputs, code revision, and weights were written to `provenance/final_pipeline_manifest.json`, which retains the exact manifest SHA-256 and bound code revision. The authorization was consumed at 2026-08-25 04:28:30 UTC and records one evaluation. No subsequent model decision used test outcomes.

ADARPro improved ROC and precision–recall curves in every endpoint (Fig. S5); exact performance is in Table S1. At frozen thresholds, false-positive rescue exceeded true-positive damage. Matched-recall counts are point estimates obtained by descriptive rethresholding on final predictions; the bootstrap distribution quantifies their pair-clustered uncertainty (Table S9). Reliability curves and lower Brier scores show that ranking gains did not sacrifice calibration (Fig. S6).

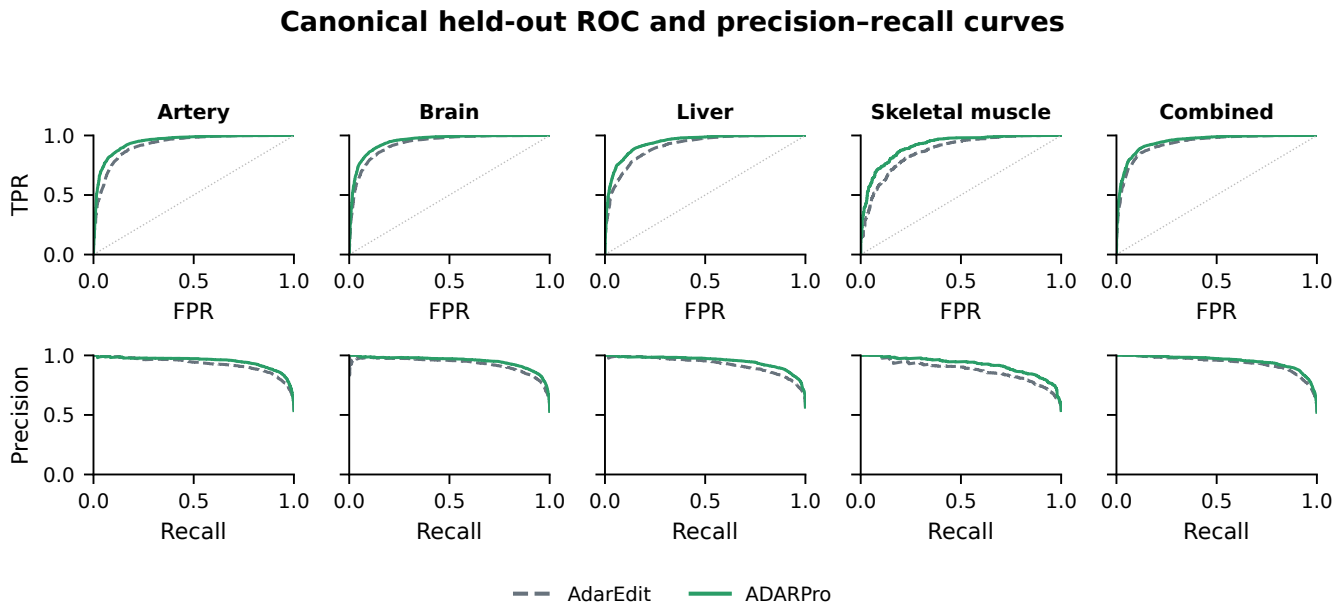


Fig. S5. Canonical held-out ROC and precision–recall curves. ADARPro and supplied-checkpoint AdarEdit are shown for four individual tissue endpoints and Combined.

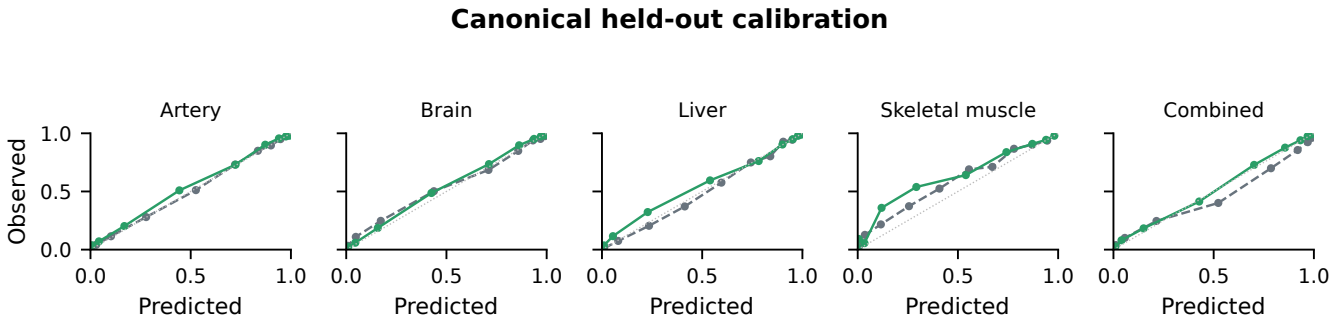


Fig. S6. Canonical held-out calibration. Reliability curves for ADARPro and supplied-checkpoint AdarEdit. Brier score is lower for ADARPro in every endpoint.

Table S9. Canonical-test false-positive transitions. Rescue and damage use frozen validation thresholds. The final column re-thresholds ADARPro to match published-baseline recall exactly.

Context	Baseline FP	FP corrected (%)	New FP	TP damaged (%)	FP avoided at matched recall
Artery	460	202 (43.9%)	42	113 (4.9%)	126
Brain	425	164 (38.6%)	55	73 (3.3%)	119
Liver	399	202 (50.6%)	43	121 (5.9%)	158
Skeletal muscle	188	83 (44.1%)	41	28 (4.3%)	68
Combined	459	147 (32.0%)	116	38 (1.7%)	148

L. Complete campaign taxonomy and provenance

Table S10 records every ranked research direction and its final disposition, including branches deferred for unavailable measurements. The immutable registry at `experiments/experiments.csv` contains one row per run with code revision, input hashes, fold definition, seed, features, hyperparameters, artifact path, serialized metrics, and test-access status. Exact OOF predictions, fold-level tables, controls, ablations, and bootstrap results remain under `experiments/runs/`; reports 00–20 preserve chronological interpretation and stopping decisions. A machine-generated numerical claim ledger maps paper quantities to authoritative artifacts and selectors. This repository-level record is more complete and useful than a page of run-family counts.

Table S10. Complete ranked performance-frontier taxonomy and final campaign decision. Full hypotheses, data sources, leakage risks, promotion rules, kill rules, and result summaries are retained in `experiments/performance_frontier_opportunities.csv`.

Rank	Axis	Direction	Final disposition
1	Biology	IRAlu architecture + transcript state	Broad panel terminated; geometry retained mechanistically
1.8	AI	Expanded high/low supervision	Selected component
2	AI	Continuous + ordinal + binary multi-task	Limited ranking signal; not selected
2.2	Biology × AI	Expanded cross-tissue shared supervision	Limited shared signal; not selected
2.5	Biology × AI	Cross-fitted all-adenosine substrate profile	Terminated after stronger-model control
3	Biology × AI	Substrate-wide multi-A + engagement latent	Terminated in tested formulation
3.4	Biology × AI	Target-centered ADAR contact geometry	Advanced to aligned local encoder
3.45	AI	Focal-node + global graph pooling	Validated fallback/complementary view
3.5	AI	Target-local multiscale CNN + graph fusion	Selected component
4	Biology	Chromatin/co-transcriptional state	Bulk peak-overlap formulation terminated
5	Biology × AI	Hierarchical engagement × site chemistry	Stopped for futility
6	Biology	In-vivo/ensemble RNA structure	Deferred: coverage/context mismatch
7	Biology × AI	Donor/sample-level multimodal model	Deferred: no matched donor outcomes
7.5	AI	Self-supervised IRAlu pretraining	Deferred: corpus not benchmark-ready
8	AI	Frozen RNA foundation-model fusion	Terminated after 150M-model confirmation
8.5	AI	Graph Transformer / GATv2 multi-scale encoder	Deferred after lower-cost local gain
9	Biology	ADAR isoform/IFN/localization state	Deferred: no matched state/outcomes
10	Biology	Cell-matched perturbation RBP effects	Deprioritized after RBP branch

Table S7 continued

Rank	Axis	Direction	Final disposition
11	AI	Calibration/stacking/ensemble	Selected final integration
11.5	AI × Biology	Unified end-to-end representation learning	Terminated after preregistered primary
12	AI	Larger/deeper GNN alone	Terminated
13	Biology	Generic RBP binding × bulk GTEx	Terminated

M. Interpretation and decisive future measurements

The evidence supports three biological levels and one learning principle. Whole-duplex sequence and structure encode intrinsic competence. Pair geometry and editing burden demonstrate a substrate-realization state, but current bulk context features cannot estimate its realized cellular value. Target/partner alignment recovers local selectivity. Pair-first sampling respects the dependence among observations from one duplex. The selected predictor integrates the two actionable representations without claiming to measure cellular environment directly.

The next high-information study should pair donor- or cell-resolved editing with transcript abundance, splicing and intron retention, ADAR1 isoform and interferon state, regulator activity, quantitative chromatin, and in-vivo RNA structure from the same samples. Perturbing ILF2/ILF3/ZFR-family regulators on substrates selected for high intrinsic competence would distinguish occupancy correlation from causal suppression. Such measurements could finally test a two-stage model in which cellular state controls substrate realization and local RNA context controls adenosine selectivity.

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